VOLTAGE REGULATOR
As Audio Amplifier

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LM338 voltage regulator can also be used as an audio amplifier. The output voltage can be controlled by varying the IC's reference voltage (pin 1), which is reflected as the output signal in the speaker. The IC can handle up to 5amp current.

Circuit and working

Circuit diagram of the audio amplifier is shown in Fig. 1. It is built around LM338 voltage regulator (IC1), BC548 transistor (T1), a few resistors (R1 through R7), two potmeters (VR1 and VR2), a loudspeaker (LS1) and a few other components.

LM338 is configured as a class-A amplifier. Initially, its output voltage is set to mid value of the maximum output voltage. On receiving an audio signal, the output signal swings to either side of the mid voltage. This audio signal is fed to the speaker through a capacitor (C3). The capacitor blocks DC components and allows only the audio signal to the speaker.

Transistor T1 controls the reference voltage of the IC as per the input signal received at its base.

The circuit works off a 12V DC, 1A current, which can be supplied by a battery or any other suitable power supply.

Test Point Voltages

Test point	Voltage
TP0	0V (GND)
TP1	Around 12V
TP2	10V

PARTS LIST

Semiconductors:

IC1	- LM338 voltage regulator
D1	- 1N4007 rectifier diode
T1	- BC548 npn transistor
LED1, LED2	- 5mm LED

Resistors (all 1/4-watt, ±5% carbon, unless stated otherwise):

R1	- 4.7-kilo-ohm
R2	- 6.8-kilo-ohm
R3	- 470-ohm
R4	- 100-ohm
R5	- 330-ohm
R6	- 2.2-kilo-ohm
R7	- 47-ohm, 5-watt
VR1	- 10-kilo-ohm potmeter
VR2	- 5-kilo-ohm potmeter

Capacitors:

C1	- 10µF, 25V electrolytic
C2	- 470µF, 25V electrolytic
C3	- 1000µF, 25V electrolytic

Miscellaneous:

BATT.1	- 12V rechargeable battery
CON1	- 2-pin connector
S1	- On/off switch
LS1	- 8-ohm, 3-watt speaker
SJ1	- Shorting jumper
	- Heat sink for IC1

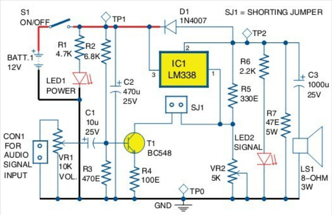

Fig. 1: Circuit diagram of the voltage regulator as audio amplifier

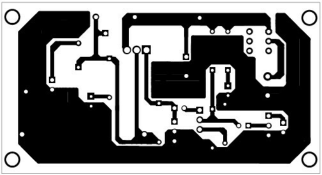

Fig. 2: Actual-size PCB layout of the voltage regulator as audio amplifier

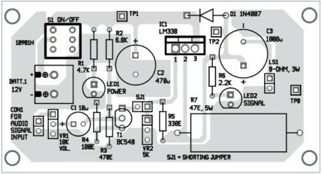

Fig. 3: Components layout for the PCB

Construction and testing

An actual-size PCB layout for the voltage regulator as audio amplifier is shown in Fig. 2 and its components layout in Fig. 3.

After assembling the components on the PCB, disconnect jumper SJ1, set VR1 to minimum and turn switch S1 'on.' Verify voltages as shown in the table.

Adjust VR2 to get about 10V DC at TP2, then connect jumper SJ1 again. Connect any audio signal as the input to CON1. The output volume can be controlled using VR1. Enclose the circuit in a suitable box. It is now ready to use. EFY

Fayaz Hassan is a manager at Visakhapatnam Steel Plant, Andhra Pradesh. He has a keen interest in microcontroller projects, mechatronics and robotics